Hallucination-Resistant Change Detection in Multimodal Large Models for Autonomous Land Management Agents

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***Abstract*—Change detection (CD) in remote sensing is essential for tracking transformations in land use, infrastructure, and landslide mapping over time. Although traditional deep learning (DL) models have shown effectiveness in standard scenarios, they often struggle with robustness and generalization, resulting in unreliable predictions in novel or complex environments. While large language models (LLMs) have been proposed to address generalization ability in various tasks, their performance is limited due to the hallucination issue. To overcome these limitations, we propose a change vector analysis (CVA)-based hallucination-resistant change detection multimodal large language model framework named CVAHR-CD-LLM, which is designed to suppress hallucinations while maintaining strong generalization capabilities. Specifically, we expand existing CD datasets, including those for landslide mapping, using a semisupervised approach and fine-tune CVAHR-CD-LLM to enhance its generalization ability. In addition, to address the hallucination issues commonly associated with LLMs, we introduce a CVA-based coordinations iterative calibration loop. Furthermore, we incorporate a self-consistency mechanism, which aggregates multiple reasoning paths to ensure robust predictions and reduce hallucinations during the CD process. This mechanism applies iterative corrections and multistep inference to refine detected change coordinates, leading to enhanced accuracy and reliability. Our method demonstrates substantial improvements in prediction precision, advancing the potential for autonomous and accurate land management applications.**

***Index Terms*—Change detection (CD), change vector analysis (CVA), hallucination reduction, iterative calibration, landslide mapping, multimodal large language models (LLMs).**

I. Introduction

CHANGE Detection (CD) is a crucial task in remote sensing, focused on identifying and analyzing changes in natural hazard evaluation [1], and environmental conditions [2] over time. Precise CD methods are crucial for a multitude of applications, ranging from landslide mapping and urban planning to disaster assessment, ecological monitoring, and land management.

The rapid advancements in satellite and aerial imagery acquisition have led to a growing demand for automated CD tools that can handle large datasets and extract meaningful insights from high-resolution images. Traditional methods, such as image differencing and postclassification comparisons, have shown limited effectiveness due to their inability to handle complex and dynamic environments. As a result, deep learning (DL) approaches have gained significant traction in CD tasks due to their superior ability to learn complex spatial and temporal patterns from data.

Traditionally, DL-based approaches, particularly those using convolutional neural networks (CNNs) [3], [4] and transformer architectures [5], [6], have been widely adopted in CD due to their capacity to learn complex features from high-resolution satellite and aerial imagery. Specifically, CNNs are effective at extracting local spatial features, while facing challenges in capturing long-frequency information. They are also prone to overfitting when trained on limited data and struggle to adapt to subtle or dynamic changes, especially in large-scale datasets or multitemporal images. Although CNN-based methods can be enhanced with preprocessing techniques, such as image alignment and segmentation, they still fail to handle contextual variations across different scenes. Transformers, originally designed for NLP, have recently gained traction in CD due to their ability to model long-range dependencies and global context through attention mechanisms. While they improve detection accuracy by capturing spatial and temporal relationships, they have limitations in local feature extraction, which is essential for identifying fine-grained changes, especially in high-resolution images or subtle variations.

Furthermore, for both CNN and transformer-based methods improving the performance of CD systems, they struggle to handle complex and dynamic environments. Challenges remain when dealing with scenes that deviate from the distribution of training data or when the detection task involves highly variable environmental conditions. These limitations can result in decreased robustness and generalization of the models, leading to suboptimal performance in real-world applications. As such, there is a growing need to explore novel approaches

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that can enhance the adaptability and generalization of CD models, which brings us to the integration of large language models (LLMs) in the context of remote sensing CD tasks.

Recent advances in LLMs offer a compelling solution to the limitations of conventional CNN and transformer frameworks. While traditional vision models primarily capture pixel-level and spatial-frequency variations, they lack cross-modal semantic reasoning and adaptability to unseen environments. In contrast, LLMs possess ultra-large parameter capacities that endow them with stronger generalization, enabling them to integrate temporal visual differences with textual semantics, such as domain prompts and contextual descriptions. This multimodal synergy allows the model to perform spatial–semantic reasoning, thereby reducing hallucination phenomena and enhancing interpretability and robustness in complex CD scenarios.

Recently, multimodal models and LLMs [7], [8] have gained traction in the AI community, demonstrating strong generalization abilities across a variety of tasks and domains. Inspired by these advancements, we propose an LLM-based framework, specifically designed to enhance generalization in CD for remote sensing applications. This ultra-large-parameter model enhances adaptability to new environments by capturing complex and unfamiliar change patterns precisely, surpassing the limitations of traditional visual feature extraction.

While integrating LLMs into the CD pipeline introduces new challenges, particularly the phenomenon of "hallucination." Hallucination refers to the generation of erroneous or irrelevant information by the model, which can severely undermine the reliability of CD results, especially in applications requiring high precision and accuracy. To overcome these limitations, we propose a CVA-based Hallucination-resistant change detection multimodal LLM framework named CVAHR-CD-LLM, effectively improving adaptability in diverse CD settings. Specifically, we introduce a CVA-based coordinations iterative calibration loop (CVA-CICL), addressing the hallucination issues commonly associated with LLMs. The calibration loop performs multistep inference, progressively correcting the coordinates of detected changes over several iterations. Furthermore, we enhance the proposed multimodal LLM framework by applying self-consistency from chain of thought (CoT), which is employed to generate multiple reasoning paths for each detection task, which are then aggregated to improve accuracy and reduce the occurrence of hallucinations. This strategy ensures more robust CD, especially in complex or unfamiliar environments where traditional models may struggle. By utilizing this approach, our framework becomes more resilient, capable of generating consistent and reliable results across a variety of challenging scenarios.

Unlike existing CLIP-based or transformer-based CD methods that primarily rely on static feature extraction or global attention, our framework introduces an iterative multimodal reasoning paradigm that jointly leverages semantic prompts and pixel-level calibration. This design enables explicit interaction between textual priors and spatial coordinates, which is absent in previous vision-only or fixed-attention models. In addition, the integration of a CVA-CICL and CoT self-consistency establishes a unique synergy between cross-modal reasoning and iterative correction, forming the core novelty of this work. An overview of the proposed CVAHR-CD-LLM framework is illustrated in Fig. 1, which shows the integration of multimodal LLM reasoning with the CVA-based iterative calibration loop for hallucination-resistant change detection.

Our contributions can be summarized as follows.

1) We propose a multimodal LLM-based framework for CD, designed to enhance the model's adaptability and generalization to novel or complex environments.
2) To mitigate the LLM's hallucination, we introduce a CVA-CICL, which applies iterative corrections to refine CD outputs over multiple inference steps.
3) Self-consistency from CoT is applied to the proposed multimodal LLM framework, generating multiple reasoning paths and aggregating them to improve accuracy and reduce hallucinations, further ensuring reliable CD in complex or unfamiliar environments.

The rest of this article is organized as follows. Section II reviews related work in CD and LLM applications in remote sensing. Section III details our proposed methodology, including the LLM architecture and CVA-based calibration loop. After that, we conduct numerical experiments in Section IV. The visualization results of the experiments are elaborated in Section V. Finally, Section VI concludes this article.

II. Related Work

Among the most widely adopted DL-based techniques for CD are CNNs and transformer architectures. In parallel with the evolution of CNN-based methods, transformer architectures have emerged as a powerful alternative for CD tasks. Originally developed for natural language processing (NLP), transformers have gained attention in RS due to their ability to model long-range dependencies and capture global contextual information [6].

A. CNN-Based Models in CD

CD in multitemporal remote sensing imagery is an essential research field with numerous applications in urban environments. Traditional CD techniques are generally divided into three main categories: image algebra methods, image transformation methods, and approaches based on DL. Image algebra techniques, including methods like image differencing and CVA, focus on quantifying changes in image intensities across temporal data [9], [10]. Image transformation methods in CD, such as Fourier transform [11] and wavelet transform [12], decompose multitemporal images into frequency or scale components, enabling the detection of changes at different levels of detail and frequencies, which is particularly effective for capturing subtle or multiscale changes in complex environments.

Recently, DL techniques have achieved significant advancements in CD by learning intricate feature representations. CNNs are commonly utilized to capture discriminative features from bitemporal imagery. Initial methods, such as Siamese CNNs [3] and multiscale Siamese networks [13], were designed to address the challenges of feature extraction specific to CD tasks. However, these models frequently lose fine-grained information

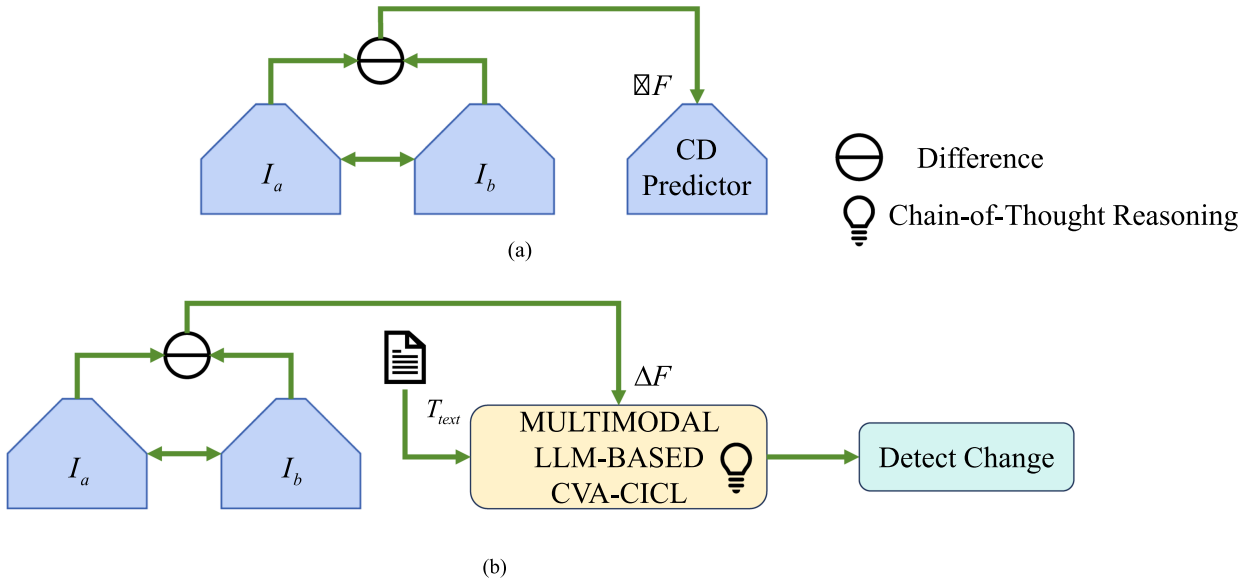


Fig. 1. CVAHR-CD-LLM framework combines CNNs and transformers' strengths while overcoming their limitations. It enhances adaptability to complex environments through LLMs, which improve generalization and accuracy. By integrating a CVA-based calibration loop and self-consistency mechanisms, our approach reduces hallucinations and ensures reliable CD in diverse scenarios.

due to the downsampling processes inherent in CNN structures. To mitigate this issue, Fang et al. [14] introduced a densely connected Siamese network that integrates multilevel features, thereby reducing semantic discrepancies and enhancing localization precision. Zhang et al. [3] introduced a triplet network-based framework for CD, where they utilized a CNN architecture to learn discriminative features for detecting subtle changes in remote sensing imagery. Their work highlighted the potential of CNNs in handling high-resolution images with varying illumination, weather conditions, and terrain types. Further advancements in CNN-based methods have been driven by the development of DL models capable of handling multitemporal data. Wang et al. [4] presented a comprehensive survey of CNN architectures for CD, discussing various approaches, such as autoencoders and Siamese networks. Their review underscored the importance of transfer learning and data augmentation techniques in addressing the challenges of limited labeled data for remote sensing tasks. They also noted the potential of CNN-based methods to achieve high performance in large-scale CD scenarios, provided that sufficient computational resources are available.

However, despite these advancements, DL-based methods, including CNNs, still face significant challenges. One of the main limitations is their inability to effectively capture global contextual information and long-range dependencies due to their localized receptive fields. This is particularly problematic in remote sensing imagery, where subtle or large-scale changes often require understanding of the broader spatial context.

B. Transformer-Based Models in CD

To address the limitations in CNN, attention mechanisms have been introduced, as they can capture long-range dependencies and global contextual information, enabling more accurate detection of subtle or large-scale changes in remote sensing imagery. Transformer-based models, which use self-attention mechanisms, are especially effective in capturing complex spatial–temporal relationships in remote sensing imagery [15], [16]. For instance, Chen et al. [5] applied a transformer-based architecture to CD, demonstrating its ability to improve accuracy and robustness in detecting changes, even in complex environments. By leveraging multiscale attention, the transformer model was able to focus on different levels of image detail, ensuring that both large-scale changes and small, subtle variations were effectively detected. Bandara et al. [6] explored the use of transformer models for high-resolution remote sensing CD. Their approach combined CNNs with transformers to achieve better performance by fusing local feature extraction capabilities of CNNs with the global contextual understanding provided by transformers. This hybrid architecture, known as a CNN-transformer network, was shown to outperform traditional CNN-based models in terms of accuracy, particularly in detecting subtle or less obvious changes in remote sensing data. Zhang et al. [15] utilized a self-attention mechanism to enhance illumination-invariant feature extraction, while others incorporated temporal-symmetric transformers for better semantic consistency [16]. Building on these ideas, Ma et al. [17] proposed an attention-regulation module to optimize the extracted difference features using attention mechanisms, which helps improve CD by reducing issues like target misdetection, false alarms, and blurry edges. Similarly, a multiscale attention module [18] that effectively captures both local and global semantic information, enhancing the accuracy of building CD by improving detail recognition and addressing unpredictable change details. However, the limited parameter capacity of CNNs and transformers restricts their ability to generalize, leading to reduced performance in unfamiliar scenarios and imprecise boundary localization.

C. Large Language Model

With the advent of large models, particularly LLMs and multimodal frameworks, there has been increasing interest in leveraging their generalization abilities for CD. Based on the ultra-large parameter scales, LLMs provide robust multimodal understanding and adaptability, allowing them to capture complex change patterns that traditional discriminative models struggle with.

Recently, researchers have started exploring the potential of LLMs in the remote sensing field [19], [20]. Zhang et al. [21] introduced RS5M, a large-scale paired dataset consisting of 5 million remote sensing images and textual descriptions, and fine-tuned a vision-language model, GeoRSCLIP, on this dataset. GeoRSCLIP demonstrated significant improvements in zero-shot classification, cross-modal retrieval, and semantic localization tasks. Similarly, Osco et al. [22] explored visual ChatGPT, an advanced LLM integrated with visual computation, to process remote sensing images. They demonstrated its capabilities in tasks, such as image description, edge detection, and segmentation, while also highlighting its limitations, such as hallucinations and domain-specific generalization issues.

Building upon these advancements, our work introduces a novel LLM-based framework specifically designed for CD tasks. Unlike previous approaches, we incorporate an iterative calibration mechanism to address hallucination issues and enhance the reliability of LLMs in diverse and challenging remote sensing scenarios. This framework not only improves the robustness of LLMs for CD but also paves the way for more accurate and interpretable CD in the field.

III. Methodology

A. System Overview

The proposed CVAHR-CD-LLM framework is a multimodal LLM-based approach designed to improve accuracy and reliability in CD for remote sensing, particularly in complex and diverse environments. This system integrates both spatial and semantic information to enhance generalization while reducing hallucinations, a common challenge in LLMs when applied to CD tasks requiring precision. At its core, CVAHR-CD-LLM combines a SigLip image encoder [23] for extracting temporal feature differences from bitemporal remote sensing images, an LLM to process multimodal inputs, and a CVA-CICL to refine predictions iteratively.

Initially, the SigLip encoder processes images from two different time points, generating feature embeddings that represent the temporal differences between the images. These feature representations, alongside corresponding textual data, are then fed into the LLM, which produces preliminary change predictions by capturing patterns across both visual and contextual modalities. The LLM operates as a multimodal model, leveraging the complementary nature of image and text features to enhance its understanding of changes occurring in the environment. This allows the model to capture complex relationships that may be difficult for traditional CNN or transformer-based models, especially in novel or unexpected change scenarios.

The textual input T_{text} in the proposed multimodal framework represents high-level semantic context that complements the visual difference features ΔF extracted by the SigLip image encoder. Each textual token sequence encodes domain-relevant prompts such as "urban expansion," "vegetation loss," or "building semantic change," together with metadata information including image acquisition date, sensor type, and scene category. These textual embeddings are jointly processed with the temporal difference map ΔF within the LLM to provide contextual guidance during reasoning, thereby enhancing the interpretability and reliability of the CD results.

To address hallucination issues and enhance spatial precision, CVAHR-CD-LLM employs a CVA-CICL. This loop iteratively corrects the change coordinates by recalibrating the model's outputs using pixel-level information from CVA, filtering out noise and refining boundaries with each pass. The CVA function compares the temporal differences between the two images at a pixel level, helping to distinguish real changes from irrelevant or false detections. This iterative process continues until the model's predictions converge, ensuring the final CD output is both accurate and robust. The iterative calibration loop ensures that the model is finely tuned to detect subtle changes in complex environments, making it suitable for applications requiring precise change localization, such as land management, urban planning, and environmental monitoring.

In addition to the aforementioned components, our framework also integrates self-consistency via CoT reasoning. CoT reasoning enables the model to generate multiple reasoning paths for each CD task, facilitating more robust predictions by aggregating diverse solutions to a single task. Specifically, CoT allows the model to break down the CD process into a series of logical steps, generating multiple candidate solutions for each step. These reasoning paths are then aggregated to provide a final prediction that incorporates the most reliable and consistent information across all generated paths. This strategy significantly enhances the model's reliability, especially in environments where change patterns may not follow well-established norms or when unexpected changes occur.

The overall system architecture as shown in Fig. 2, integrates these components into a unified pipeline, where the image encoder extracts meaningful temporal features, the LLM processes multimodal inputs to generate preliminary predictions, and the iterative calibration loop refines the results to ensure spatial accuracy. The integration of self-consistency through CoT reasoning provides an additional layer of robustness, improving the model's ability to generalize and maintain high accuracy in unfamiliar or challenging environments.

To provide an intuitive understanding of the multimodal reasoning process inside the proposed framework, Fig. 3 expands the LLM module in Fig. 2, illustrating how textual and visual information interact within the reasoning and calibration pipeline. As illustrated in Fig. 3, the LLM fusion core integrates the temporal difference feature ΔF generated by the SigLip encoder with the textual semantic embedding T_{text}. Within the LLM, cross-attention layers align visual and textual tokens, while the embedded CoT reasoning produces multiple reasoning paths (r_1, r_2, r_3) aggregated through a self-consistency strategy

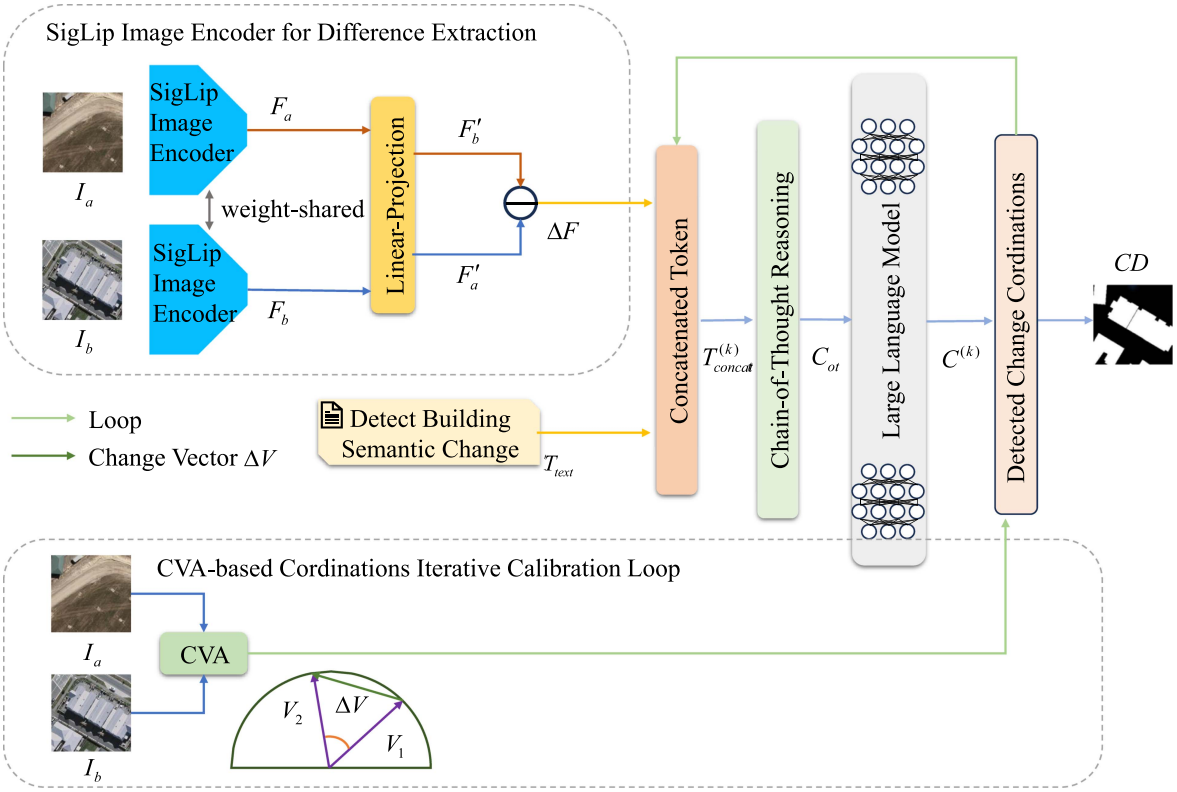


Fig. 2. Architecture of the proposed CVAHR-CD-LLM framework for CD in remote sensing. The system incorporates a SigLip image encoder to extract temporal feature differences from bitemporal images, an LLM for multimodal reasoning, and a CICL to iteratively refine predictions using pixel-level spatial information. The framework also leverages self-consistency through CoT reasoning, which enhances robustness by generating and aggregating multiple reasoning paths, enabling precise and generalizable CD in complex and diverse environments.

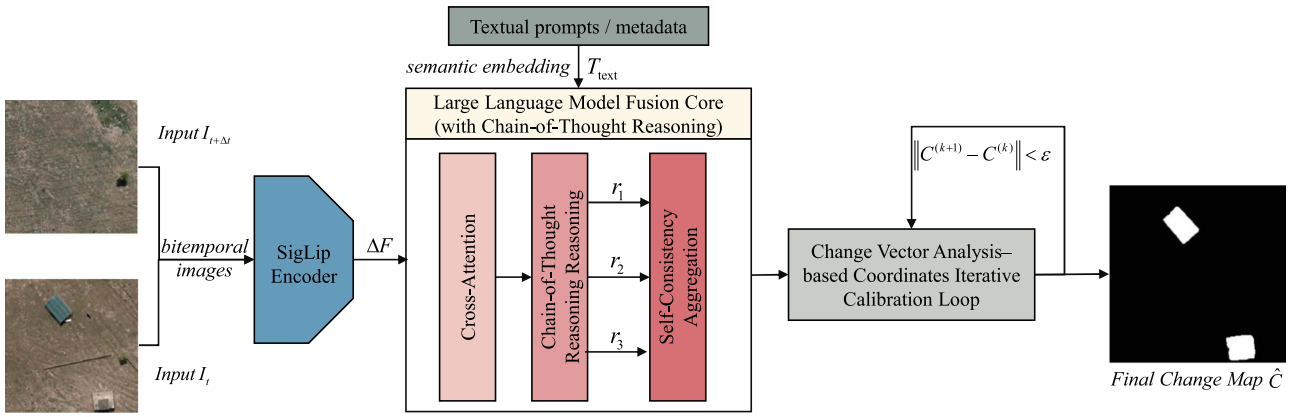


Fig. 3. Multimodal reasoning workflow within the LLM fusion core of the proposed CVAHR-CD-LLM. The SigLip-derived difference features ΔF are fused with textual semantics T_{text} via cross-attention, followed by CoT reasoning that generates multiple reasoning paths (r_1, r_2, r_3). A self-consistency module aggregates these paths to produce a stable coarse change map, which is further refined by the CVA-CICL.

to ensure stable inference. The coarse change map is then iteratively refined by the CVA-CICL, which enforces the convergence condition $\|C^{(k+1)} - C^{(k)}\| < \varepsilon$, yielding the final change result $\hat{C}$.

B. Model Architecture

The architecture of the CVAHR-CD-LLM framework is designed to integrate spatial, temporal, and semantic information for robust CD in remote sensing applications. The model consists of three key components: the SigLip image encoder for temporal difference extraction, the CVA-CICL for refining change predictions, and the integration of self-consistency via CoT reasoning to enhance model decision-making. These components are seamlessly combined to address the challenges of detecting changes in complex environments while minimizing hallucinations and improving spatial accuracy.

1) SigLip Image Encoder for Difference Extraction: The SigLip image encoder processes bitemporal remote sensing images I_a and I_b, which are remote sensing images taken at

two distinct time points t_a and t_b, respectively. First, the two images I_a and I_b are passed through the encoder network ϕ_{shared}, parameterized by θ to produce the feature representations

$$F_a = \phi_{\text{shared}}(I_a; \theta), \quad F_b = \phi_{\text{shared}}(I_b; \theta) \tag{1}$$

where $F_a \in \mathbb{R}^{H\times W\times D}$ and $F_b \in \mathbb{R}^{H\times W\times D}$ represent the feature maps of the images I_a and I_b, D is the feature space dimensionality. These feature maps are then projected into a shared latent space using a projection head ψ

$$F'_a = \psi(F_a; \phi), \quad F'_b = \psi(F_b; \phi) \tag{2}$$

where $F'_a, F'_b \in \mathbb{R}^{H\times W\times D'}$, and D' is the dimensionality of the latent space. To obtain a concise representation of temporal differences, we compute the normalized temporal difference map ΔF

$$\Delta F_{ij} = \frac{F'_{b,ij} - F'_{a,ij}}{\sqrt{\|F'_{a,ij}\|^2 + \|F'_{b,ij}\|^2 + \epsilon}} \tag{3}$$

where i, j represent spatial coordinates, and $\epsilon > 0$ is a small constant added for numerical stability. The resulting $\Delta F \in \mathbb{R}^{H\times W\times D'}$ captures both the magnitude and direction of temporal changes in the latent feature space. The difference vector ΔF forms the core input for the subsequent components of the framework. This difference vector encapsulates the essential transformations between the two images, allowing for the identification of temporal changes, such as urbanization or deforestation, while suppressing background noise.

Unlike conventional models like CLIP [24], which utilize a softmax objective for cross-modal similarity computation, SigLip adopts a sigmoid-based similarity scoring approach to better handle independent image-text pairs. The similarity score z_{ij} between image i and text j is calculated as

$$\mathcal{S} = \frac{1}{|B|}\sum_{i=1}^{|B|}\sum_{j=1}^{|B|}\log\left(1 + e^{z_{ij}\cdot(-t_i\cdot y_j + b)}\right) \tag{4}$$

where z_{ij} represents the similarity score, t_i and y_j denote the latent space embeddings of image i and text j, respectively, and b is a trainable bias term. This sigmoid-based formulation avoids the need for complex summations across the entire batch, enabling more efficient processing, which is particularly advantageous for large-scale datasets commonly encountered in remote sensing applications. Such datasets often involve high-resolution and multimodal data, requiring scalable and computationally efficient approaches.

In this multimodal framework, the SigLip encoder processes each image individually to produce feature representations that emphasize temporal distinctions rather than static attributes. The resultant difference ΔF, combined with textual embeddings T_{text}, serves as the primary input to the subsequent stages of the model, including the CVA-based calibration loop, focusing the model on detecting meaningful changes rather than unrelated background features. This modular and robust difference extraction enables the system to maintain accuracy and adaptability across varied environmental settings.

2) Large Language Model: The temporal difference map ΔF is then combined with the textual context T_{text}, encoded as a sequence of tokens $\{w_1, w_2, \ldots, w_L\}$, where L is the sequence length. Together, the multimodal input representation is expressed as

$$T_{\text{LLM}} = \text{concat}(\Delta F, T_{\text{text}}) \tag{5}$$

which serves as input to the LLM. The LLM generates an initial coarse change map C_0

$$C_0 = \text{LLM}(T_{\text{LLM}}) \tag{6}$$

where $C_0 \in \mathbb{R}^{H\times W}$ provides a preliminary estimate of the change likelihood at each pixel. However, this coarse prediction often contains spatial inaccuracies, which are addressed through an iterative refinement process.

3) CVA-Based Iterative Calibration Loop (CVA-CICL): The CVA-CICL is a core component of the CVAHR-CD-LLM framework, designed to iteratively refine the detected change coordinates, reducing hallucinations and enhancing spatial accuracy in CD. This loop integrates semantic information from the LLM with precise pixel-level calibration from CVA, achieving a balance between the model's generalization capabilities and the spatial precision required for accurate remote sensing applications.

The loop operates in discrete iterations, indexed by k. In each iteration, an updated token $T^{(k)}_{\text{concat}}$ is created by concatenating three key components, as follows:

1) the difference vector ΔF generated by the SigLip image encoder, which encodes the temporal changes between images I_a and I_b;
2) the text token T_{text}, providing contextual information relevant to the type of change (e.g., "building semantic change");
3) the previously calibrated change coordinates $P^{(k-1)}_{\text{CVA}}$, representing refined pixel-level change locations from the previous iteration.

The refinement process begins with the initialization of the change map $C^{(0)}$

$$C^{(0)} = C_0 \tag{7}$$

where C_0 is the coarse prediction generated by the LLM in the initial stage. At each iteration k, the model constructs a concatenated input representation that combines the temporal difference map ΔF, the textual context T_{text}, and the calibrated change map from the previous iteration $P^{(k-1)}_{\text{CVA}}$. The concatenated input at iteration k is given by

$$T^{(k)}_{\text{concat}} = [\Delta F, T_{\text{text}}, P^{(k-1)}_{\text{CVA}}] \tag{8}$$

where ΔF is the normalized temporal feature difference, T_{text} is the encoded textual context, and $P^{(k-1)}_{\text{CVA}}$ is the pixel-level calibration mask obtained from the previous iteration. This multimodal input is fed into the LLM to produce an updated change map

$$C^{(k)} = \text{LLM}(T^{(k)}_{\text{concat}}). \tag{9}$$

The iterative process allows the model to correct errors in the initial prediction by effectively integrating pixel-level calibration information into the change map. The updated prediction $C^{(k)}$ reflects the model's current understanding of changes between I_a and I_b, informed by both the encoded differences ΔF and the contextual text input.

At each iteration, a pixel-level calibration mask $P_{\text{CVA}}^{(k)}$ is computed using CVA. Specifically, for each pixel (i, j), the magnitude of the change vector is calculated as

$$P_{\text{CVA},ij}^{(k)} = \|I_{b,ij} - I_{a,ij}\| \tag{10}$$

where $I_a, I_b \in \mathbb{R}^{H\times W\times C}$ are the two input images, and $I_{a,ij}$ and $I_{b,ij}$ represent the pixel values at location (i, j) in the respective images. The calibration mask $P_{\text{CVA}}^{(k)}$ captures the intensity of pixel-wise differences, which is subsequently used to adjust the predicted change map $C^{(k)}$. This adjustment mechanism ensures that the refinement process focuses on meaningful temporal changes while mitigating the influence of noise or irrelevant features. The iterative refinement process continues until the change map stabilizes, i.e., when

$$\|C^{(k+1)} - C^{(k)}\| < \epsilon \tag{11}$$

where ϵ is a predefined convergence threshold. This criterion ensures that the iterative process terminates when successive change maps exhibit minimal differences, indicating that the model has reached a stable and refined prediction.

In our implementation, the iterative calibration process described above typically converges within $k = 3$–5 iterations under the threshold condition in (11), ensuring stable refinement of the change coordinates without excessive computational overhead. Each iteration updates the change map $C^{(k)}$ by re-evaluating the concatenated input $T_{\text{concat}}^{(k)}$ defined in (8), where the convergence criterion $\|C^{(k+1)} - C^{(k)}\| < \varepsilon$ is empirically satisfied after a few iterations on all tested datasets (CDD, GVLM, and WHU-CD). The iterative updates introduce only a moderate runtime increase—approximately proportional to the number of refinement steps—while providing significant gains in spatial accuracy and stability. This balance between computational cost and accuracy demonstrates the efficiency of the proposed CVA-CICL mechanism in achieving high-fidelity CD.

By setting this convergence criterion, the loop ensures that the model stabilizes on a consistent and reliable set of change coordinates, eliminating spurious predictions. The final output of the loop is a robust CD map that combines the semantic understanding of the LLM with the spatial accuracy of CVA, effectively reducing hallucinations and improving precision. This iterative calibration mechanism enables the CVAHR-CD-LLM framework to deliver high-fidelity CD results, suitable for applications that demand high spatial and semantic accuracy, such as land management, urban planning, and environmental monitoring.

4) Self-Consistency Via CoT Reasoning: In the domain of CD for remote sensing applications, uncertainty is an inherent challenge due to factors, such as sensor noise, variations in environmental conditions, and potential distortions in the images. To address these challenges and enhance the reliability of predictions, we integrate CoT reasoning into our framework, as shown in Fig. 4. CoT reasoning introduces a structured process where the model not only generates a final prediction but also explores multiple intermediate steps or "thoughts" that lead to the final conclusion. This structured reasoning is particularly important for handling the complexity and ambiguity often present in remote sensing data, where different hypotheses might explain the observed changes.

The core idea of CoT reasoning is to decompose the prediction task into a series of logical steps, where each step corresponds to an intermediate reasoning process or hypothesis. By incrementally refining its understanding, the model generates multiple reasoning paths that explore different possible explanations for the observed changes in bitemporal images. Each reasoning path evaluates temporal differences, semantic context, and competing hypotheses, enabling the model to account for complex environmental factors and reduce ambiguity. This structured approach ensures that the most consistent and plausible explanations are prioritized, enhancing robustness against noise, improving generalization to unseen environments, and minimizing false positives (FP). The resulting aggregation of these reasoning paths produces a more accurate and reliable prediction, addressing the inherent challenges of dynamic and noisy remote sensing scenarios.

To enhance the robustness of the model's decision-making, the CVAHR-CD-LLM framework incorporates CoT reasoning. CoT encourages the model to generate multiple reasoning paths for each CD task, thereby allowing it to explore diverse hypotheses rather than relying on a single reasoning path. This results in a more accurate and reliable final prediction by reducing bias and overfitting. CoT self-consistency works by iterating through multiple reasoning steps. For each CD task, the model generates a series of intermediate logical steps or substeps that guide the final prediction. This iterative process leads to a set of candidate reasoning paths $\mathcal{R} = \{r_1, r_2, \ldots, r_n\}$, where each r_i associated with a distinct set of assumptions and intermediate conclusions

$$r_i = \text{CoTReasoning}(C_0, \Delta F, \mathcal{P}_i). \tag{12}$$

Here, $\mathcal{P}_i$ is the set of logical assumptions or conditions for the ith reasoning path. Each reasoning path generates an intermediate prediction $f(r_i)$

$$f(r_i) = \text{Prediction}(r_i). \tag{13}$$

For each reasoning path r_i, an intermediate CD map $f(r_i)$ is produced. The final CD map C_{final} is obtained by aggregating the intermediate predictions from all reasoning paths. This aggregation process ensures that the most consistent and confident predictions across all reasoning paths are prioritized, which is crucial for reducing uncertainties in remote sensing data. The CoT reasoning process can be formalized as follows:

$$C_{ot} = \text{agg}\left([f(r_1), f(r_2), \ldots, f(r_n)]\right) \tag{14}$$

where $f(r_i)$ is the prediction from reasoning path r_i and $\text{agg}(\cdot)$ represents the aggregation function, such as weighted averaging

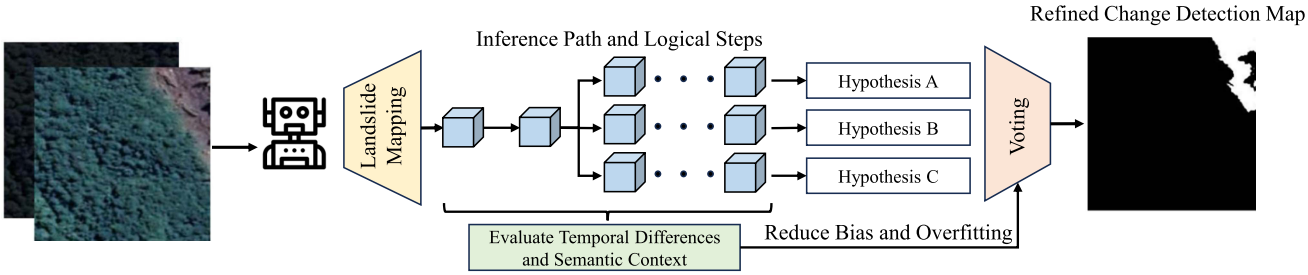


Fig. 4. Proposed framework integrating CoT reasoning for improved CD. The CoT module generates multiple reasoning paths to explore different hypotheses, and aggregates them to produce robust predictions.

and voting. This method enhances the model's ability to handle noise, ambiguity, or incomplete data by exploring multiple hypotheses in parallel.

The aggregation function $\text{agg}(\cdot)$ can be defined as a weighted average, where w_i denotes the weight of each reasoning path based on its confidence

$$\text{agg}(\{f(r_1), f(r_2), \ldots, f(r_n)\}) = \sum_{i=1}^{n} w_i f(r_i). \tag{15}$$

This ensures that more confident predictions are given greater influence in the final result. To Further improve CD precision, the CVAHR-CD-LLM framework incorporates a coordinations iterative calibration loop (CVA). After generating the difference vector ΔF and performing CoT reasoning, the output is refined in each iteration through the CVA loop. In each cycle, the difference vector ΔF, combined with contextual semantic information T_{text}, and the previous change prediction $P_{\text{CVA}}^{(k-1)}$, are used to update the model's understanding of the spatial changes. The updated token $T_{\text{concat}}^{(k)}$ is then input into the LLM, which outputs a refined set of change coordinates $C^{(k)}$.

CoT reasoning provides several distinct advantages in CD tasks. First, it effectively addresses the inherent uncertainty in remote sensing data, such as noise or ambiguities in observed changes. By generating multiple hypotheses and evaluating them through aggregation, the model ensures that the most consistent and plausible explanation is selected. For instance, if one path attributes a detected change to urbanization and another considers seasonal vegetation variations, the aggregation step reconciles these hypotheses, reducing the risk of overfitting to irrelevant features. This capability enables the model to account for complex environmental factors that may influence the observed changes.

In addition to handling uncertainty, CoT reasoning significantly improves the model's robustness. By exploring diverse hypotheses simultaneously, the model reduces the likelihood of misinterpreting background noise or spurious patterns as genuine changes. Each reasoning path is designed to explore different assumptions or interpretations, refining predictions at intermediate steps and allowing the model to generalize better to new and unseen environments. This is particularly important in remote sensing, where data quality can vary due to atmospheric interference, sensor limitations, or uneven illumination conditions.

Another key advantage of CoT reasoning is its ability to minimize hallucinations, which refer to FP or irrelevant changes often caused by overgeneralization or noise. By generating multiple reasoning paths, CoT reasoning minimizes the impact of any single path that may produce a hallucination. The aggregation process ensures that such spurious predictions are outweighed by more consistent and plausible paths, ultimately improving the overall accuracy (OA) of the model. This approach is especially valuable in complex tasks like urbanization detection, where the model must distinguish between genuine changes, such as new construction, and unrelated phenomena, such as seasonal vegetation dynamics.

By iterating through hypothesis generation and refinement, CoT reasoning provides a structured framework for reasoning about changes. Each reasoning path undergoes intermediate refinements, incorporating additional features, semantic context, and prior knowledge to improve its predictions. The final CD output is derived by aggregating the results from these paths, with more weight given to the most consistent and confident hypotheses. This iterative and structured reasoning makes CoT reasoning a powerful tool for achieving reliable and accurate CD in dynamic and noisy remote sensing environments.

This section presents an evaluation of the proposed CVAHR-CD-LLM framework on three widely-used remote sensing CD datasets: CDD [29], GVLM [30], and WHU-CD [31]. The evaluation is organized as follows.

C. Datasets and Preprocessing

In this section, we describe the datasets used in our experiments, which include CDD, GVLM, and WHU-CD. These datasets provide diverse remote sensing data for evaluating CD methods, encompassing a variety of geographic regions, image resolutions, and types of annotated changes.

The CDD dataset contains high-resolution optical satellite images with a spatial resolution of 0.3 m. It covers a wide range of environments, from urban to rural areas, and includes 500 pairs of multitemporal images. The changes annotated in this dataset primarily relate to urbanization, land-use alterations, and infrastructure development.

The GVLM dataset is focused on urban CD and includes 622 pairs of high-resolution optical images (0.5 m) from multiple Chinese cities. It annotates changes related to urban expansion, building construction, and infrastructure developments. The WHU-CD dataset contains both optical and radar images with resolutions of 2.5 m and 5 m, respectively. This dataset includes 34 image pairs and focuses on land-use changes, such as agriculture, deforestation, and urbanization, primarily in rural and

agricultural regions of China. To ensure the models trained on these datasets perform effectively, several preprocessing techniques are employed. Image normalization is applied to standardize pixel values using min–max or Z-score normalization, ensuring consistent input ranges across all images. Image resizing is performed to reduce computational overhead, resizing images to 256×256 or 512×512 pixels while preserving essential spatial details. In addition, data augmentation techniques, such as random rotation, flipping, cropping, color jittering, and noise injection, are used to enhance model robustness and prevent overfitting. These augmentations simulate various real-world conditions, such as changes in viewpoint, lighting, and sensor noise.

For optical imagery, radiometric normalization and color augmentation are applied to stabilize illumination variations and enhance RGB-semantic fidelity. For radar data, speckle noise is suppressed through adaptive filtering and polarization normalization before being fed into the SigLip encoder. During the CVA-CICL refinement, radar sequences typically require two to four additional iterations to achieve convergence, improving robustness under adverse weather and low-visibility conditions.

D. Evaluation Metrics

To assess the performance of our proposed CVAHR-CD-LLM framework, we utilize several standard evaluation metrics commonly used in CD tasks. These metrics provide a detailed analysis of the model's accuracy and effectiveness in detecting meaningful changes in remote sensing data.

OA is a fundamental metric that measures the proportion of correctly classified pixels across all categories, including both changed and unchanged regions. It is calculated as

$$\text{OA} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \tag{16}$$

where true positives (TP) represents the number of correctly detected changes, true negatives (TN) represents the correctly detected unchanged areas, FP denotes the incorrectly detected changes, and false negatives (FN) represents the incorrectly classified unchanged areas.

Intersection over union (IoU) evaluates the spatial overlap between predicted and true change areas. This metric is crucial for assessing how well the detected changes match the actual changes on the ground. The IoU is computed as

$$\text{IoU} = \frac{|A_{\text{pred}} \cap A_{\text{true}}|}{|A_{\text{pred}} \cup A_{\text{true}}|} \tag{17}$$

where A_{pred} is the predicted change area and A_{true} is the true change area. A higher IoU indicates a better spatial match between the predicted and true change regions.

F1-score provides a balanced measure of a model's precision and recall, offering a single metric that considers both FP and FN. The $F1$-score is computed as the harmonic mean of precision and recall

$$F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{18}$$

where *precision* is defined as

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{19}$$

and *recall* is defined as

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}. \tag{20}$$

The $F1$-score is particularly valuable for imbalanced datasets, as it ensures that both FP and FN are considered in the evaluation.

E. Experimental Setup

In this section, we describe the experimental configuration, including hardware specifications, software libraries, and the hyperparameters used for training our CVAHR-CD-LLM framework. We also detail the implementation specifics of the CVA-CICL. All experiments were conducted on a system with four NVIDIA A100 GPUs, which provided the necessary computational power for training large multimodal models. This hardware setup enabled efficient parallel processing of high-resolution remote sensing images, ensuring scalability and fast convergence during training. For hyperparameters, the batch size was set to 16 to ensure efficient processing and avoid GPU memory overload. The learning rate was initialized at 1×10^{-4}, with a linear warm-up over the first 10% of the total training steps, followed by a cosine decay to improve generalization and prevent overfitting. The training process was conducted over 50 epochs, which provided sufficient time for the model to converge without excessive computation time. The AdamW optimizer was used with a weight decay of 0.01 to prevent overfitting. Gradient clipping was applied with a value of 5.0 to stabilize training and avoid exploding gradients. The training time for each dataset varied based on its size and complexity.

IV. Experimental Visualization Results

In this section, we present the visualization results of our experiments to demonstrate the effectiveness of the proposed CVAHR-CD-LLM framework on different CD datasets. Fig. 5 presents qualitative comparisons on the CDD dataset, where the proposed CVAHR-CD-LLM produces cleaner change maps with fewer spurious responses compared with existing methods. Fig. 6 further illustrates qualitative results on the WHU-CD dataset, showing that CVAHR-CD-LLM achieves more accurate change boundaries and effectively suppresses false positives in complex scenes.

We visualize the predicted CD results on GVLM. The visual results highlight the model's ability to detect both large-scale and subtle changes, reduce hallucinations, and generalize to previously unseen environments. The GVLM dataset contains high-resolution images of both urban and rural areas, with a focus on land-use changes, including construction, vegetation change, and urban sprawl. Fig. 7 presents the CD results on the GVLM dataset. The column a and b display the input images, the column c shows the ground truth annotations, and the column d–h shows the models' predictions. Our framework is able to detect a wide range of changes, from large-scale urbanization to smaller-scale vegetation shifts. Particularly notable is the

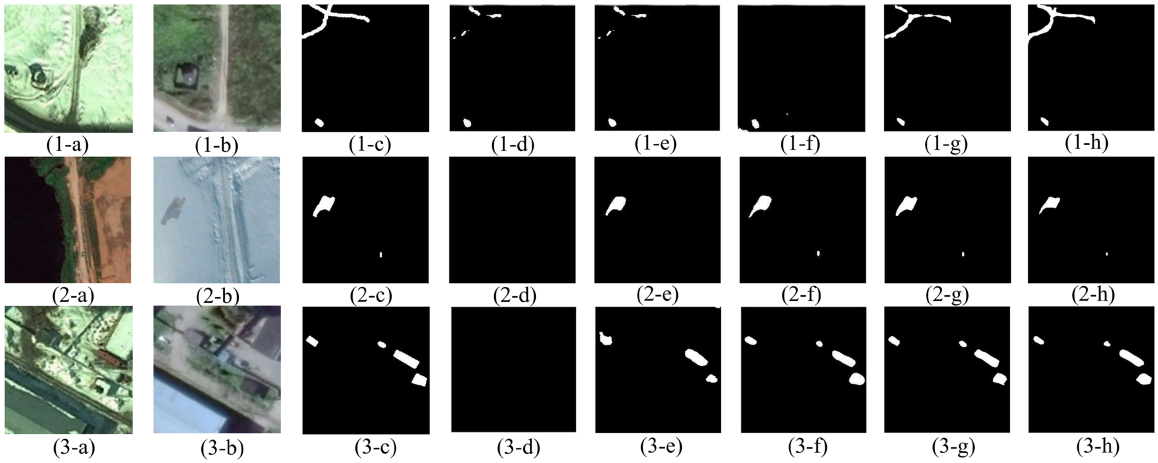


Fig. 5. Comparison of different state-of-the-art CD methods on CDD. (a) Prechange image. (b) Postchange image. (c) Ground-truth. (d) FC-SC [25]. (e) SNUNet [26]. (f) DT-SCN [27]. (g) AMTNet [28]. (h) CVAHR-CD-LLM (ours).

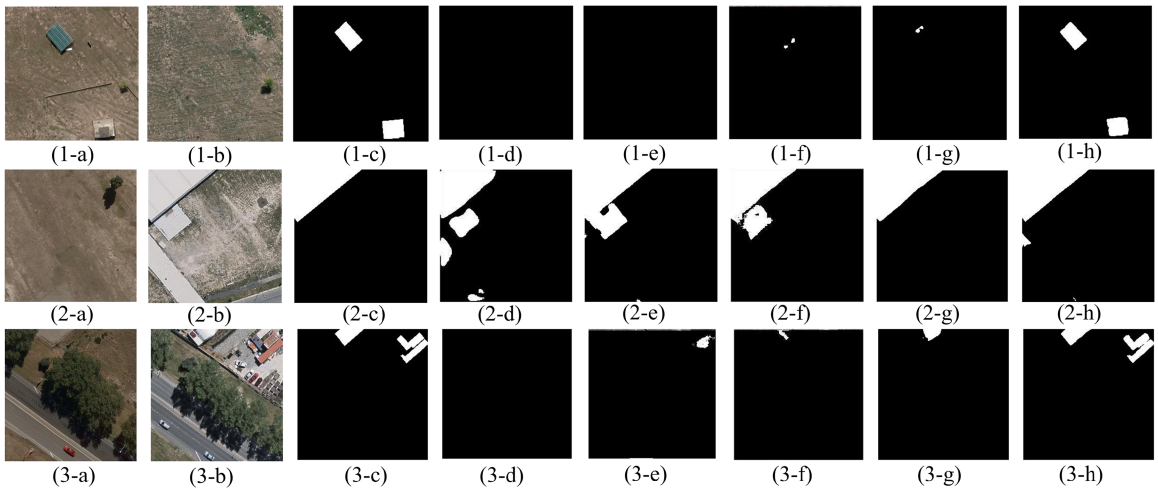


Fig. 6. Comparison of different state-of-the-art CD methods on WHU. (a) Prechange image. (b) Postchange image. (c) Ground-truth. (d) FC-SC [25]. (e) SNUNet [26]. (f) DT-SCN [27]. (g) AMTNet [28]. (h) CVAHR-CD-LLM (ours).

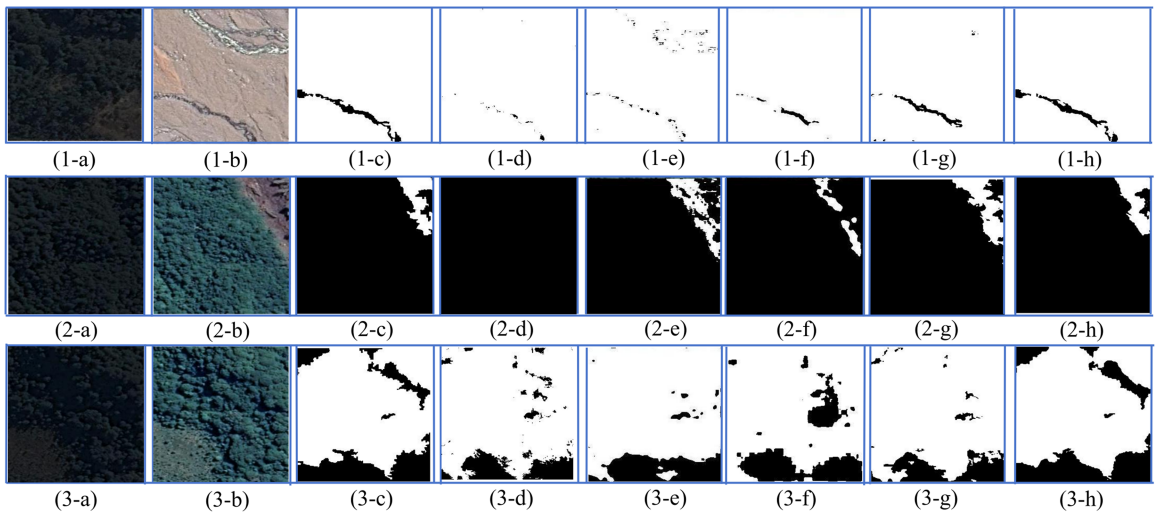


Fig. 7. Comparison of different state-of-the-art CD methods on GVLM. (a) Prechange image. (b) Postchange image. (c) Ground-truth. (d) FC-SC [25]. (e) SNUNet [26]. (f) DT-SCN [27]. (g) AMTNet [28]. (h) CVAHR-CD-LLM (ours).

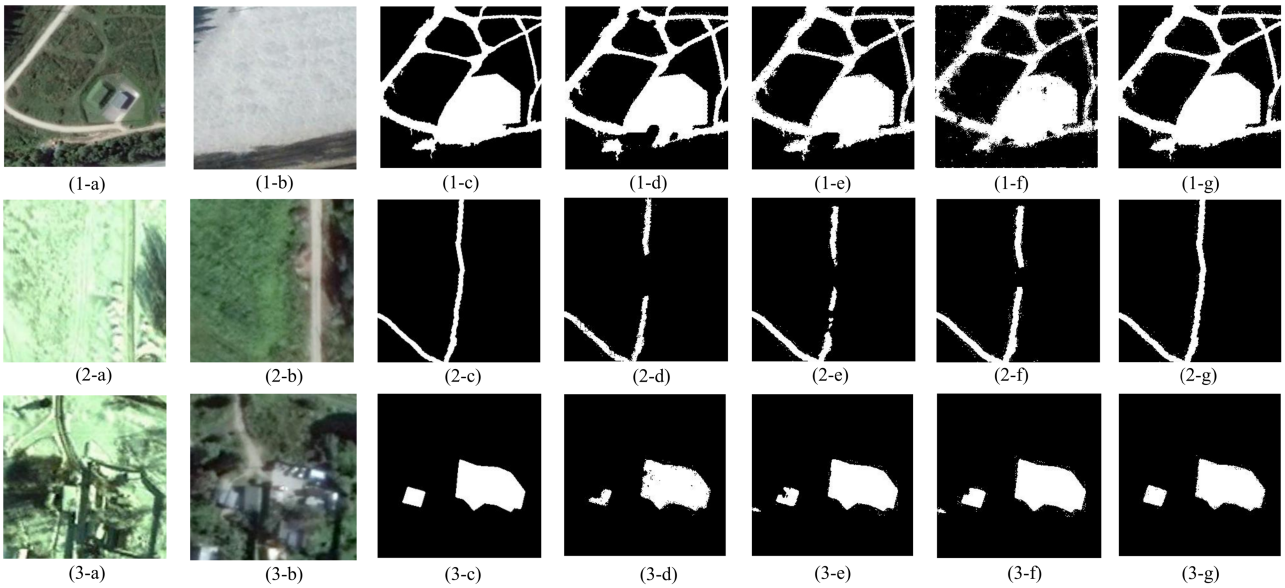


Fig. 8. (a) Prechange image. (b) Postchange image. (c) Ground-truth. (d) Baseline model: Performance of the model without the CVA-CICL and self-consistency components. (e) With self-consistency only: Performance when self-consistency is applied but CVA-CICL is removed. (f) With CVA-CICL Only: Performance when CVA-CICL is used but self-consistency is removed. (g) Proposed model: Performance with both CVA-CICL and self-consistency components integrated. These configurations highlight the contributions of each component to overall performance improvement in CD tasks.

TABLE I
ABLATION EXPERIMENT RESULTS(%) ON THE GVLM DATASET

CVA-CICL	Self-Consistency	OA	F1	IoU
		97.16	89.11	84.56
	✓	98.45	92.84	86.72
✓		98.97	93.51	88.14
✓	✓	**99.39**	**95.36**	**90.02**

The values in bold are the best.

model's ability to accurately detect changes in rural areas where subtle land modifications can often be overlooked. The visual results demonstrate the model's robustness in handling both urban and rural environments with varying landscape characteristics.

As illustrated in Fig. 7, the proposed CVAHR-CD-LLM produces noticeably sharper building boundaries and clearer delineation of vegetation regions compared with other state-of-the-art methods. In dense urban scenes with shadowed structures or reflective surfaces, our framework effectively suppresses hallucinated FP, leading to visibly cleaner predictions. Moreover, small-scale structures, such as minor buildings (10–20 pixels) are more accurately localized, achieving an IoU of 90.02%, compared with 85%–88%for the best baseline approaches. These observations verify that the CVA-CICL refines spatial coordinates through iterative calibration, while the self-consistency mechanism stabilizes multimodal reasoning, jointly enhancing spatial precision and reliability in high-resolution, complex environments.

These results clearly illustrate the strengths of our CVAHR-CD-LLM framework across three distinct datasets. On each dataset, the proposed model successfully identifies both large-scale and minor changes with high accuracy, reduces hallucinations, and generalizes well to unseen environments, making it a robust solution for real-world CD tasks in remote sensing applications.

A. Ablation Studies

To understand the contributions of each component in the CVAHR-CD-LLM framework, we conducted a series of ablation studies. The goal is to isolate the effects of critical components, such as the CVA-CICL and self-consistency. These components were selectively removed or modified, and the performance of each variant was evaluated on the GVLM dataset.

The results, presented in Table I, show that each individual component contributes significantly to the overall performance of the proposed method.

Baseline model: The model without any additional components (i.e., no CVA-CICL and no self-consistency) achieves an OA of 97.16%, $F1$ score of 89.11%, and IoU of 84.56%. While these results are decent, they indicate that the baseline model suffers from limitations in robustness and hallucination resistance, which is particularly evident in the relatively lower $F1$ score and IoU. As shown in Fig. 8(d), the baseline model detects significant changes but struggles with subtle or complex changes, often resulting in noisy or incomplete detection, as indicated by the scattered and incomplete highlighted areas in the visual output. These results highlight the model's limitations in robustness and hallucination resistance.

Without CVA-CICL, with self-consistency: When only self-consistency is applied (i.e., without CVA-CICL), the performance improves slightly, reaching an OA of 98.45%, $F1$ score of 92.84%, and IoU of 86.72%. As illustrated in Fig. 8(e), the model is more capable of handling complex changes, showing less noise in the detection output. The aggregated reasoning paths contribute to a more accurate detection of changes, but

TABLE II
QUANTITATIVE EXPERIMENTAL RESULTS (%) ON THE CDD

Method	Recall	Precision	OA	F1	IoU
FC-SC [25]	71.10	78.62	94.55	74.67	58.87
SNUNet [26]	80.29	84.52	95.73	82.35	69.91
DT-SCN [27]	89.54	92.76	97.95	91.12	83.59
AMTNet [28]	93.87	**94.96**	98.68	94.41	89.37
CVAHR-CD-LLM	**96.19**	94.49	**98.90**	**95.33**	**91.21**

The values in bold are the best and the underlined values indicate the second-best performance for each evaluation metric.

TABLE III
QUANTITATIVE EXPERIMENTAL RESULTS (%) ON THE WHU-CD

Method	Recall	Precision	OA	F1	IoU
FC-SC [25]	86.54	72.03	98.42	78.62	64.37
SNUNet [26]	81.33	85.66	98.68	83.44	71.39
DT-SCN [27]	93.60	88.05	99.32	90.74	83.55
AMTNet [28]	91.26	92.46	99.35	91.86	84.63
CVAHR-CD-LLM	**93.58**	**94.69**	**99.56**	**94.13**	**88.69**

The values in bold are the best.

there are still instances of minor misclassification, particularly in regions with subtle changes, highlighting that self-consistency alone is insufficient to fully resolve these challenges.

With CVA-CICL, without self-consistency: In this configuration, where the *CVA-CICL* is employed but self-consistency is not used, the model achieves an OA of 98.97%, an $F1$ score of 93.51%, and an IoU of 88.14%. The results in Fig. 8(f) show that the iterative correction process via CVA-CICL significantly refines the detection of subtle changes, reducing errors and improving OA. However, while the output is cleaner and more precise than the previous configurations, there are still some occasional detection artifacts, particularly in challenging regions where CD is subtle or context-dependent.

With both CVA-CICL and self-consistency (proposed model): Finally, when both CVA-CICL and self-consistency are combined, the model achieves the highest performance with an OA of 99.39%, $F1$ score of 95.36%, and IoU of 90.02%. The visual results in Fig. 8(g) show that this configuration achieves near-perfect detection performance, with precise delineation of both large and subtle changes. The iterative calibration of CVA-CICL works synergistically with the multipath aggregation of self-consistency, resulting in the most accurate and robust detection across all regions. This combination ensures minimal noise and misclassification, demonstrating the full potential of the integrated approach.

These ablation results demonstrate that both the CVA-CICL and self-consistency play essential roles in improving the overall performance of the CD model. The full integration of both components enables the CVAHR-CD-LLM framework to achieve substantial gains in accuracy, robustness, and hallucination resistance, making it well-suited for real-world applications where CD tasks are often complex and varied. The combination of these techniques represents a novel and powerful approach to addressing the challenges associated with large-scale remote sensing CD.

B. Qualitative Analysis

The experimental results presented in Tables II–IV provide valuable insights into the performance of various CD methods.

TABLE IV
QUANTITATIVE EXPERIMENTAL RESULTS (%) ON THE GVLM

Method	Recall	Precision	OA	F1	IoU
FC-SC [25]	89.53	74.47	96.47	81.31	67.24
SNUNet [26]	91.67	87.22	98.97	89.39	80.85
DT-SCN [27]	92.95	82.05	97.16	87.16	77.31
AMTNet [28]	90.23	93.70	90.05	91.93	85.03
CVAHR-CD-LLM	**95.11**	**95.61**	**99.39**	**95.36**	**90.02**

The values in bold are the best.

In all three datasets, the proposed CVAHR-CD-LLM framework consistently outperforms other state-of-the-art methods in terms of key evaluation metrics, such as recall, precision, OA, $F1$ score, and IoU.

On the CDD dataset, as shown in Table II, the proposed framework achieves significant improvements across all metrics. Specifically, our method reaches a recall of 96.19%, a precision of 94.49%, an OA of 98.90%, an $F1$ score of 95.33%, and an IoU of 91.21%. This demonstrates not only superior performance in detecting changes but also a high level of precision in identifying relevant areas of change. The model's ability to maintain high performance, particularly in the $F1$ score, highlights its robustness in balancing both recall and precision.

For the WHU-CD dataset, shown in Table III, our approach again leads in all performance metrics. The recall reaches 93.58%, with a precision of 94.69%, OA of 99.56%, $F1$ of 94.13%, and IoU of 88.69%. The OA value further emphasizes the reliability of the proposed method, suggesting that it can be effectively applied to real-world, large-scale remote sensing applications with varying environmental conditions.

The GVLM dataset, presented in Table IV, illustrates similar superior performance, with the proposed model achieving a recall of 95.11%, precision of 95.61%, OA of 99.39%, $F1$ of 95.36%, and IoU of 90.02%. These results underscore the adaptability of CVAHR-CD-LLM in diverse remote sensing environments and its ability to achieve high precision even in more complex scenarios.

In summary, the proposed method consistently outperforms other techniques, such as FC-SC, SNUNet, DT-SCN, and AMTNet, which demonstrates the efficacy of our multimodal LLM-based framework in enhancing generalization and robustness. The significant improvements in IoU and $F1$ score further highlight the model's ability to reduce FP and FN, providing more reliable and precise CD outputs. The qualitative performance analysis demonstrates that the CVAHR-CD-LLM framework offers superior CD results across a variety of datasets, showcasing its potential for real-world applications in remote sensing. These improvements can be attributed to the novel integration of multimodal LLMs, iterative calibration, and self-consistency mechanisms, which together enhance both the model's generalization ability and its robustness against challenges, such as hallucination and environment variability.

To further clarify how the proposed framework addresses the inherent limitations of previous CD approaches, we provide a deeper interpretation of the processing process and the resulting improvements. Conventional CNN-based methods rely heavily on local convolutions and, therefore, struggle to maintain consistency when illumination or texture varies across time, while transformer-based models, though capable of global attention,

often blur structural boundaries and amplify noise due to coarse-grained fusion. In contrast, the proposed CVAHR-CD-LLM performs a progressive reasoning workflow: first, the SigLip encoder extracts bitemporal difference features ΔF capturing low-level spatial discrepancies; second, the LLM interprets these visual cues together with textual context T_{text} through multimodal attention, establishing semantic understanding of potential changes; finally, the CVA-CICL refines the preliminary map by aligning semantic reasoning with pixel-level coordinates until convergence under the criterion $\|C^{(k+1)} - C^{(k)}\| < \varepsilon$. This reasoning-to-calibration process effectively mitigates fragmented detections, reduces hallucinated FP, and preserves fine object boundaries—especially in shadowed or high-reflectance regions. The consistent gains observed in Tables II–IV, thus, stem from this synergistic pipeline, which combines semantic inference with spatial correction to overcome the weaknesses of vision-only architectures.

V. Discussion

This section discusses why the proposed CVAHR-CD-LLM consistently improves hallucination suppression, boundary preservation, and cross-scene generalization in remote-sensing CD. Rather than relying on architectural complexity or additional supervision, these gains stem from the explicit interaction between multimodal semantic reasoning and iterative spatial calibration. Building upon this perspective, the following discussion analyzes the internal mechanisms underlying these performance gains, contrasts the proposed framework with existing paradigms, and examines its practical implications and future extensions.

A. Mechanistic Insights and Cross-Module Synergy

The observed improvement can be attributed to the coordinated operation of two core components: the CVA-CICL and the self-consistency mechanism embedded within the multimodal LLM. The CVA-CICL module performs progressive refinement between the difference feature ΔF extracted from the SigLip encoder and the textual embedding T_{text}, ensuring better semantic-spatial alignment across iterations. This iterative calibration substantially reduces spatial misalignments and hallucinated responses in areas with high texture complexity or distributional shifts. Meanwhile, the self-consistency mechanism aggregates multiple reasoning paths to achieve stable consensus inference, mitigating stochastic errors and semantic ambiguity during CoT reasoning. This synergistic interaction between calibration and consensus allows CVAHR-CD-LLM to retain both high spatial precision and semantic interpretability, effectively addressing fundamental challenges in multimodal reasoning that conventional vision-only networks struggle with.

B. Comparison With Existing Paradigms

From a methodological perspective, most existing CNN-based approaches are vulnerable to local appearance variations, while transformer-based models often trade boundary precision for global attention. The proposed framework departs from both paradigms by decoupling high-level semantic reasoning from pixel-level spatial calibration. When contrasted with CNN and transformer-based CD methods, CVAHR-CD-LLM demonstrates superior robustness under heterogeneous imaging conditions. CNNs are constrained by local receptive fields and limited global context modeling, while transformers capture long-range dependencies but tend to blur fine structures due to coarse attention fusion. In contrast, the proposed framework fuses visual–textual priors through the LLM backbone and iteratively refines pixel-level changes via CVA-CICL. This dual process ensures reliable detection across diverse landscapes, including urban expansion and vegetative degradation. Unlike previous models that rely on fixed feature hierarchies, CVAHR-CD-LLM dynamically reasons over multimodal inputs, leveraging large-scale linguistic knowledge to infer cross-modal relationships. Such design yields consistent generalization across datasets, such as CDD, GVLM, and WHU-CD, where the framework achieves balanced improvements in both global reasoning and boundary precision without additional retraining.

C. Tradeoffs, Constraints, and Computational Aspects

The consistent gains observed on CDD, GVLM, and WHU-CD are not merely attributed to stronger feature representations, but to the progressive reasoning-to-calibration pipeline. Semantic cues guide the LLM to identify plausible change hypotheses, while CVA-CICL iteratively corrects spatial inconsistencies, preventing fragmented detections and hallucinated FP.

Despite its advantages, the framework entails several operational tradeoffs. The iterative calibration loop, while enhancing accuracy, introduces additional inference latency proportional to the number of iterations. However, convergence is typically achieved within 3–5 steps under the condition

$$\|C^{(k+1)} - C^{(k)}\| < \varepsilon$$

ensuring efficiency in practice. Furthermore, the semisupervised data augmentation strategy used to enhance generalization may inadvertently propagate mild domain bias if the synthesized samples fail to represent rare atmospheric or topographic anomalies. Residual hallucinations may still appear under extreme multimodal inconsistency, such as severe radar-optical misalignment. These aspects delineate potential directions where scalability, data diversity, and environmental robustness warrant deeper investigation.

D. Practical Implications for Autonomous Land Management

The architectural innovations of CVAHR-CD-LLM hold substantial implications for autonomous land management and other geospatial intelligence systems. By linking textual prompts (e.g., “urban expansion,” “vegetation change”) with observed spatial differences, the framework generates semantically interpretable change maps that support transparent decision-making. Its hallucination-resistant characteristics ensure that change products remain reliable under complex environmental variability, thereby strengthening trust in automated Earth observation pipelines. Such robustness is particularly beneficial for real-world applications, including rapid postdisaster assessment,

ecological restoration monitoring, and long-term urban development planning. In this sense, CVAHR-CD-LLM bridges high-level reasoning from LLMs with precise geospatial analytics, contributing to a new class of intelligent agents capable of scalable and explainable environmental management.

E. Prospects for Enhancement

Future efforts will focus on improving efficiency and generalization while maintaining interpretability. Hybridization with complementary modalities—such as incorporating radar polarization and hyperspectral cues—could further increase robustness under adverse conditions. Lightweight optimization techniques, including parameter-efficient adaptation, early-stopping criteria, and knowledge distillation, may alleviate computational costs in edge deployments. In addition, longitudinal experiments on multiregional datasets can assess scalability, while domain-adaptive fine-tuning strategies will enhance cross-regional transferability. Integrating external knowledge bases and adaptive prompt engineering into the LLM may also reinforce semantic consistency and further mitigate hallucinations in open-world scenarios.

The proposed CVAHR-CD-LLM achieves a balance between accuracy, interpretability, and computational feasibility by combining multimodal reasoning with iterative spatial calibration. The framework not only mitigates key limitations of existing CD models but also lays a foundation for scalable, trustworthy, and semantically guided remote-sensing analytics—an essential step toward the next generation of autonomous land-management systems.

VI. Conclusion

This article introduces CVAHR-CD-LLM, a multimodal framework for CD in remote sensing, designed to improve generalization and robustness by integrating LLMs with CVA. The combination of the SigLip image encoder and an iterative CVA-based calibration loop allows the model to reduce hallucinations and enhance spatial accuracy. Furthermore, the incorporation of a self-consistency mechanism ensures that multiple reasoning paths are aggregated to produce more robust predictions, further reducing the likelihood of hallucinations and improving model reliability. Experiments on CDD, GVLM, and WHU-CD demonstrate that CVAHR-CD-LLM outperforms traditional methods in accuracy and reliability, proving effective in complex, real-world scenarios. Future work will focus on extending this framework to other data types, optimizing the calibration process for greater efficiency, and exploring further refinements to the self-consistency mechanism for even stronger CD performance. Overall, CVAHR-CD-LLM establishes a new paradigm for hallucination-resistant multimodal reasoning in remote sensing, paving the way for scalable, real-time applications in disaster response, sustainable urban planning, and autonomous land-management systems.

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